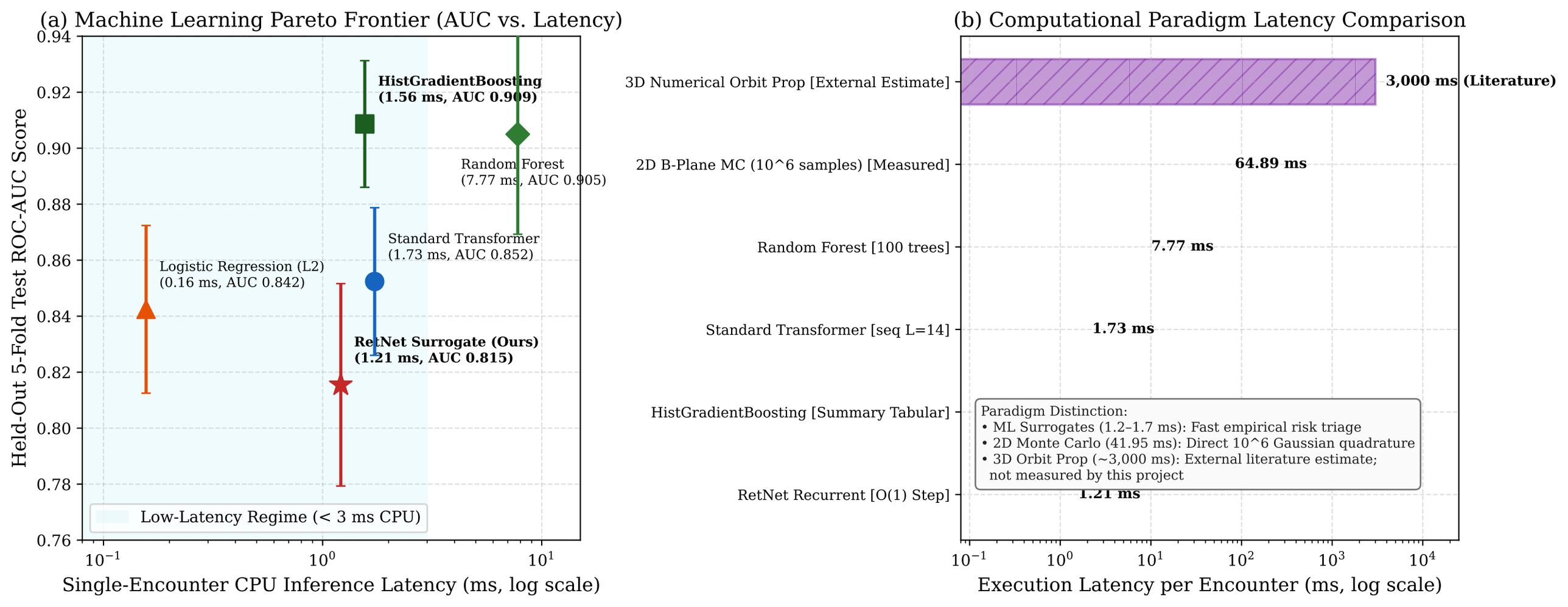
(a) Machine Learning Pareto Frontier (AUC vs. Latency)
Held-Out 5-Fold Test ROC-AUC Score
Single-Encounter CPU Inference Latency (ms, log scale)
HistGradientBoosting
(1.56 ms, AUC 0.909)
Random Forest
(7.77 ms, AUC 0.905)
Standard Transformer
(1.73 ms, AUC 0.852)
Logistic Regression (L2)
(0.16 ms, AUC 0.842)
RetNet Surrogate (Ours)
(1.21 ms, AUC 0.815)
Low-Latency Regime (< 3 ms CPU)
(b) Computational Paradigm Latency Comparison
3D Numerical Orbit Prop [External Estimate]
3,000 ms (Literature)
2D B-Plane MC (10^6 samples) [Measured]
64.89 ms
Random Forest [100 trees]
7.77 ms
Standard Transformer [seq L=14]
1.73 ms
HistGradientBoosting [Summary Tabular]
RetNet Recurrent [O(1) Step]
1.21 ms
Paradigm Distinction:
• ML Surrogates (1.2–1.7 ms): Fast empirical risk triage
• 2D Monte Carlo (41.95 ms): Direct 10^6 Gaussian quadrature
• 3D Orbit Prop (~3,000 ms): External literature estimate;
not measured by this project
Execution Latency per Encounter (ms, log scale)